

Automated Integration and Intelligent Harmonization of Multi-Source Geospatial Data for Urban Land Record Management

KSHETRA — research summary and evidence base

Team NovaX · Smart India Hackathon 2026 · Problem Statement 26013

This document sets out what the published literature and the programme record establish, what they leave unsolved, and which of those open problems our prototype addresses. Claims are graded: **[V]** verified against a primary or peer-reviewed source, **[S]** secondary — consistently reported but not traced to the original, **[U]** unverified. Figures quoted for KSHETRA were measured by a harness in the repository and are inserted into this document at build time.

1. Why Indian land records are hard — the structural cause

The fragmentation is not accidental and it is not recent. The Survey of India was established in 1767 and conducted revenue survey nationally until **1904**; after that date **each state became responsible for its own cadastral survey and evolved its own system** [10] **[V]**. Every downstream interoperability problem — divergent vocabulary, divergent scales, divergent legal primacy — descends from that single administrative decision.

Two consequences matter technically.

Legal primacy differs between north and south

Northern states recognise the graphic map as the legal document; southern states prioritise the **Field Measurement Book** [11] **[V]**. In the north the polygon carries authority; in the south the *measurements* carry it and the polygon is a derived sketch. A single national harmonisation policy is therefore legally wrong, and any system that silently rewrites geometry is destroying the record in one half of the country while merely correcting it in the other.

Vocabulary differs structurally, not just lexically

South Indian survey numbers carry a recursive sub-division grammar ($123/1A$); north Indian khasra numbering conventions vary by settlement. The record of rights is the *khatauni* in UP, the *jamabandi* in Punjab and Haryana, the *patta / chitta* in Tamil Nadu. And “**khata**” **denotes a revenue holding account in the north but a municipal property-tax account in urban Karnataka** — a live hazard in an urban project. Schema matching must therefore be structure-aware, not merely synonym-aware.

2. The problem, quantified

2.1 Scale of the urban gap

India has **7,933+ urban settlements** covering roughly **10.2 million hectares**, and **4,912 urban local bodies** [9, 12] **[V]**. Only **four states** maintain structured urban land records — a figure corroborated independently by DoLR's own NAKSHA booklet, which names Tamil Nadu, Maharashtra, Gujarat and Goa [12] **[V]**. Approximately 13 million urban households live in 108,000 slums [9] **[V]**.

Under DILRMP, **49.10%** of villages hold geo-referenced cadastral maps [13] **[V]**.

2.2 Economic and legal consequence

Several widely-circulated “land dispute” statistics in India are recycled with drifting attribution. Only traceable figures appear below; the ones we deliberately do not use are listed in Appendix A.

Claim

Source

Grade

Land market distortions erode annual GDP growth by an estimated 1.3%	Satyarthi et al., Frontiers in Sustainable Cities, 2026 [9]	[V]
ULBs generate only 32% of revenue internally, on sub-optimal tax assessment	Satyarthi et al., 2026 [9]	[V]
Land and property taxes are 0.6% of GDP in low-income economies vs 2.2% in industrialised ones	Satyarthi et al., 2026 [9]	[V]
Property disputes comprise 60–70% of Indian civil litigation	Satyarthi et al., 2026 [9]	[V] as a peer-reviewed statement
Roughly two-thirds of civil matters surveyed concerned land and property	DAKSH Access to Justice Survey 2015-16; 9,329 litigants, 305 locations, 24 states [14]	[V] as a survey of litigants, not a census
Cadastral resurvey rate of ₹56,725 per sq km	Tamil Nadu Dept. of Survey & Settlement, DILRMP review, 2024 [15]	[V]
Resurvey of a district tranche takes 2–3 years even with drone + DGPS	Tamil Nadu, 2024 [15]	[V]

2.3 Positional accuracy of legacy cadastral maps — the central measurement

Sengupta, Lemmen, Devos, Bandyopadhyay and van der Veen georeferenced **310 analogue cadastral sheets** at 1:3960 covering a 327 km² planning area in West Bengal — 258 mouzas and 26 municipal wards, mapped mostly from pre-1920s surveys and 1950s work — against GeoEye-1 pan-sharpened imagery using 10–15 ground control points per sheet and a first-order transformation [5] [V].

RMSE range	Sheets
< 2.00 m	7
2.01 – 3.00 m	64
3.01 – 4.00 m	185
4.01 – 5.00 m	47
> 5.01 m	7

The modal positional error of an Indian legacy cadastral sheet is **3–4 metres**, and over 88% of sheets fall between 2 and 5 metres [5].

The same study compared transformation orders on a sample of sheets:

Sample	1st order (m)	2nd order (m)	Reduction
1	3.668	3.177	13%
2	3.171	2.510	21%
3	3.962	1.339	66%
4	4.089	3.096	24%
5	5.010	2.889	42%

A higher-order transformation cuts error by up to two thirds on the worst sheets — but **choosing the transformation order per sheet is currently a human judgement** [5]. Automating that selection from the residual structure of each sheet is a concrete, demonstrable contribution, and it is what our escalating transform selector does.

Documented failure modes in legacy sheets [5] [V]

- **Gaps and overlaps between adjoining mouza sheets** — adjacent sheets do not fit; they overlap or leave voids.
- **Multi-sheet mouzas** with unclear division between sheets.

- **Canal double-counting** — a canal used as a shared boundary is drawn on both adjacent sheets, so the same area is counted twice.
- **River dynamics** — area lost to or accreted from a river, giving large legal-versus-digitised discrepancies.
- **Paper degradation** — shrinkage, wrinkling, folding, tearing; scanning adds further error.
- **Crude map lines** — the scanned line is often more than 4–5 pixels wide, so boundary ambiguity of decimetres to metres exists before any transformation is applied.
- **Area attribute inconsistency** — digitised parcel area does not equal the authorised area recorded in the RoR.

Our damage model reproduces five of these seven directly (misregistration, smooth warp, vertex jitter, sliver and overlap injection, recorded-versus-geometric area drift). That the damage model is derived from a documented failure taxonomy rather than invented is why we consider the recovery figures meaningful.

2.4 What the new data will look like

Survey of India technical circular T-260/1147-Project, dated 10 February 2025, specifies for NAKSHA: orthorectified imagery at 5 cm GSD with **RMSE(x,y) ≤ 10 cm**; DSM/DTM at 0.5 m spacing with **RMSE(z) ≤ 15 cm**; CORS control better than 5 cm; 70%/60% overlap; a minimum of four corner and one centre GCP. The coordinate reference system is settled: **UTM projection on WGS 84**, with vertical referenced to the Indian Vertical Datum via the Sol geoid model [6] [V].

3. The accuracy mismatch that defines the problem

New NAKSHA orthoimagery: $RMSE \leq 10$ cm [6]. Legacy cadastral sheet: modal $RMSE$ 3–4 m [5]. That is a **30–40× accuracy mismatch between the two layers the problem statement asks us to harmonise**.

Naive overlay is therefore not merely imprecise; it is meaningless. Urban plots are of the order of 11 m across, so a legacy parcel displaced by 5–15 m overlaps its *neighbour* more than it overlaps its own counterpart. This single comparison determines the architecture: georeferencing must precede matching, and any system that matches first is measuring the wrong pair.

We observed this directly. Matching before georeferencing gave a blocking recall of 0.58 — 42% of true correspondences never reached the model — and the model compensated by learning to join on khasra number instead of geometry. Reordering the pipeline took blocking recall to **1.0**.

4. What the literature establishes

4.1 The state of the field — Chen, Nazeer, Lee & Wong [1]

A systematic review of AI in cadastre published March 2026. Its conclusions: deep learning has removed the manual-interpretation bottleneck in boundary extraction, using CNNs and Transformers for pixel-level semantic segmentation; natural-language methods now extract structure from paper archives; and deep models detect parcel change and support integrated spatial/non-spatial analysis [1] [V].

It then states what remains open:

“...challenges remain, including differences in multi-temporal data processing, spatial semantic ambiguity, and the lack of large-scale, high-quality annotated data. Future research can focus on improving model generalization, advancing cross-modal data fusion, and providing recommendations for the development of a reliable and practical intelligent cadastral system.” [1]

Three of those four define this project: the absence of annotated ground truth, cross-modal fusion, and generalisation beyond the area a method was tuned on.

4.2 Alignment under noisy supervision — Girard, Charpiat & Tarabalka [2]

The closest published statement of our core problem. The authors align misregistered cadastral polygons to imagery where no correct annotation exists, using a multi-resolution U-Net that predicts a 2-D displacement field,

trained over repeated rounds that re-correct the annotations between rounds. Alignment error fell by more than a factor of three by round two; round three added nothing [2] [V].

Two findings bear directly on our method.

Their training data is built by applying known random deformations and learning to invert them — formally, a dataset $D = \{(I, J_rand, f_rand)\}$ where the deformation is generated and therefore known [2]. This is the same principle as our damage harness, which is why we treat manufactured ground truth as an established method rather than an improvisation.

They report two honest limitations we inherit. A perfect alignment score is unattainable because the reference annotations are themselves ambiguous — many buildings are outlined by a coarse polygon, making best-alignment ill-posed with multiple equally good solutions. And their model learns only *smooth* displacement fields, so adjoining buildings requiring a discontinuous correction fail [2].

Our Gaussian-process error field carries the same smoothness assumption. We treat it as a stated limitation, and it is the reason our uncertainty model reports an irreducible floor — 0.75 m in the evaluated ward — that no quantity of survey control removes.

4.3 Cadastral parcel adjustment with SAM and ICP — Suwardhi et al. [3]

The closest operational system. UAV orthophotos at 5 cm GSD are segmented by the Segment Anything Model without manual prompting; displacement vectors come from ICP alignment; adjustment proceeds hierarchically from rigid block (LS1) to scaled block (LS2) to individual parcel (LS3). Evaluated over two urban villages in Cimahi, Indonesia — Karangmekar with 81 blocks and 3,227 parcels, and Baros with 96 blocks and 2,971 parcels, selected for contrasting block structure and data quality. Implemented client-server with a Python REST API [3] [V].

What it leaves open is the instructive part. Because no correct geometry exists for those areas either, the authors evaluate *by proxy*: they count how many times cadastral polygons split the SAM-derived boundary segments, and treat fewer splits as better alignment [3]. That measures internal consistency, not accuracy. It is the state of the art, and it cannot report its error in metres.

The work is also single-source, and does not address attribute schema inference, CRS/datum/unit heterogeneity, tenure semantics, calibrated confidence, or where to survey next.

4.4 Boundary revision by deep learning — Fetai, Grigillo & Lisec [4]

A modified CNN detects visible land boundaries from image-based mapping and uses them to revise existing cadastral data [4] [V]. This is the upstream of our pipeline, and we deliberately do not rebuild it: extraction is well served, and the problem statement asks for integration of extracted features rather than for another extractor.

4.5 Method imported from outside the domain

Our survey-planning component rests on sensor placement in Gaussian processes. The objective — total reduction in posterior variance over a set of observation locations — is monotone submodular, so greedy selection is within a factor $(1 - 1/e)$ of optimal and the exact problem is NP-hard [8] [V]. The result is standard in machine learning; we found no application of it to cadastral ground-control planning.

5. What the existing tools do, and do not do

Tool	What it genuinely does	What it does not do
GRASS v.clean	Topology repair: break/clean polygons from non-topological formats, remove sub-threshold areas by dissolving into the neighbour with the longest shared boundary, fuzzy snapping, vertex pruning	A human chooses the snapping and area thresholds. Wrong threshold means destroyed parcels or surviving slivers. It is a tool, not a decision-maker.
PostGIS Topology	Persistent topological model with shared edges and faces; prevents gaps and overlaps by construction going forward	Getting legacy dirty geometry <i>into</i> a valid topology is the hard part; common practice is to clean in GRASS first.

QGIS + GDAL/PROJ	Georeferencer with polynomial 1st–3rd order, thin-plate spline, projective and Helmert; topology checker; reprojection	All manual, per layer, per operator. GCP selection by hand. No cross-source reasoning.
FME / Esri Data Interoperability	Industry-standard spatial ETL; hundreds of formats; visual schema mapping	The mapping is declared by a human, not inferred. FME will execute a mapping faithfully; it will not tell you that <code>KHASRA_NO</code> and <code>SY_NO</code> are the same concept, nor that they disagree for 12% of parcels. Also proprietary, at 4,912-ULB scale.
Esri Parcel Fabric	Mature cadastral editing and QA	Deed-based COGO assumptions; not built for khasra / FMB / mouza semantics or a 30–40× inter-layer accuracy mismatch.
SAM, Mask R-CNN, U-Net	Strong at extracting <i>visible</i> boundaries from imagery	Cadastral boundaries are frequently invisible — no wall, no fence, no hedge. This is the ceiling of the pure-vision approach.
Bhu-Naksha (the Indian incumbent)	National cadastral map software in use across many states	Manually entered per-state scale factors on shapefile import — UP ×4000, Himachal ×22 where <i>karam</i> was digitised in centimetres — and hand-written per-state RoR adapters. The incumbent national solution to unit and schema heterogeneity is a hard-coded constant. [V]

6. What is not solved

1. **No automated, evidence-weighted reconciliation across accuracy tiers.** Every tool assumes you already know which layer is right. Nothing decides that here the 10 cm orthoimagery wins, there the legal recorded area wins, and elsewhere the case must be adjudicated by a human.
2. **Transformation-model selection remains a human judgement.** Second-order beat first-order by up to 66% on the worst sheets [5], but choosing per sheet is manual. No system selects the transformation model per sheet from its own residual structure.
3. **Schema matching for Indian land attributes is entirely manual.** FME and Esri require a human to draw every mapping; Bhu-Naksha's answer is bespoke per-state adapters. We found no published automated schema-matching system for Indian land-record attributes.
4. **Topology cleaning has no legal-consequence awareness.** `v.clean rmbarea` will dissolve a 3 m² sliver — which may be a real, owned, taxed, litigated parcel. No tool distinguishes a digitisation artefact from a genuine tiny urban parcel, though the distinction is classifiable from area, compactness, presence of a khasra number, presence in the RoR, and presence of a building footprint.
5. **Vision models find visible boundaries; cadastres contain invisible ones.** Fusing imagery evidence, legacy geometry, textual record and utility topology to infer an invisible boundary is not addressed in the literature we found.
6. **Cross-agency entity resolution is unaddressed.** The NAKSHA progress review names “parallel, non-interoperable records” across Revenue, ULB, Development Authority and Sub-Registrar as a top friction point, lists insufficient cross-agency interoperability as a systemic void, and recommends *automated GeoAI-driven updating mechanisms* [9] [V].
7. **The state of the art is single-source and single-country.** Suwardhi et al. align one cadastral layer to one UAV orthophoto [3]. Extending block-adjustment to an n-source, schema-aware, CRS-resolving, legally-aware pipeline is what the problem statement actually asks for.
8. **No open benchmark or ground truth exists for Indian cadastral harmonisation.** Producing even a small one for a pilot ULB would be a defensible deliverable in itself.

7. Standards and the canonical target

ISO 19152 — Land Administration Domain Model. Original edition 2012; current ISO 19152-1:2024 generic conceptual model, with Part 2 on land registration in development. LADM is explicitly *not* a data product specification: its stated purpose is “not to replace existing systems, but rather to provide a formal language for describing them.” Four core packages: parties; basic administrative units carrying rights, restrictions and responsibilities; spatial units; and spatial sources and representations [16] [V].

Sengupta et al. asked in 2013 how to convert colonial maps and records into an LADM-based database, **how to document and publish the geometric quality of existing maps**, and **how to integrate more accurate data after re-survey** [17] [V]. That last question — fusing a new high-accuracy survey with an existing low-accuracy legal record without destroying the legal record — is this problem statement, posed in the literature thirteen years ago and still open.

Adopting LADM as the canonical target schema lets us say that every source is mapped into an ISO-conformant model rather than into one we invented.

ULPIN. A 14-character parcel identifier generated from geo-coordinates under DILRMP. The mechanism is documented; the exact character composition rule is [U] and we do not reproduce it. Our implementation is a faithful *shape*, labelled as such in code and interface.

8. Our approach

The components are not individually novel and we do not claim them. Segmentation, coordinate transformation, topology repair, change detection and human review all exist. The contribution lies in three places, each answering one of the gaps named above.

8.1 Manufactured ground truth — accuracy measured, not asserted

Both [2] and [3] are limited by the same thing: no correct geometry exists to score against, so one reports relative improvement and the other a proxy. We generate a clean synthetic cadastre and damage a copy through eleven stages whose parameters are recorded, then score recovery against the key. Each stage reproduces a failure mode documented in [5]:

Damage stage	Real-world cause
Similarity misregistration	sheet georeferenced from too few, poorly spread GCPs
Smooth non-linear warp	paper shrinkage, scanner distortion, map-sheet join
Vertex jitter	manual digitisation from a paper sheet
Vertex decimation	generalisation during digitisation
Sliver and gap injection	adjoining sheets digitised independently
Overlap injection	shared boundaries double-counted between epochs
Self-intersection	careless digitising, unclosed rings
Attribute schema drift	every department names its columns differently
Owner-name corruption	transliteration variance across romanisations
Area unit drift	records kept in bigha and biswa, not m ²
Block renumbering; split, merge, deletion	resurvey and subsequent mutation

This addresses the review's “lack of large-scale, high-quality annotated data” [1] by constructing it, following the deformation-injection principle of [2].

8.2 Calibrated confidence rather than a weighted score

A weighted sum of similarity terms is not a probability and cannot be checked against anything. Candidate pairs are scored by gradient-boosted trees over 29 features — geometric agreement, shape descriptors, attribute similarity after transliteration folding, and context features describing how ambiguous each candidate is relative to its alternatives — then mapped through isotonic regression fitted on held-out data, so a stated probability corresponds to an observed frequency. Expected calibration error is **0.00009**.

Calibration is what makes an explicit triage threshold possible instead of an arbitrary cut-off, and it is the difference between a number that decorates the interface and one that decides what a human looks at.

8.3 Closing the loop to the field

The NAKSHA progress review identifies ground truthing as the programme's most delayed component, with 55 of roughly 150 pilot ULBs below 60% completion [9]. Because every parcel carries a calibrated uncertainty, the system fits a Gaussian process to the residual displacement field, then selects survey locations by greedy submodular maximisation of total variance reduction [8], and reports the expected improvement.

The recommendation is then **validated rather than asserted**: the selected control is actually applied, georeferencing is re-run, and achieved error is compared against both the prediction and against random placement of the same number of points.

8.4 Ordering, and why it is fixed

ingest → schema inference → validate → georeference → blocking → match → assign → topology
→ conflict → change → uncertainty → targeting → serve

Georeferencing precedes matching for the reason set out in section 3. Conflict resolution applies inverse-variance weighting to measured quantities and legal authority to recorded ones — GNSS control wins on geometry, the revenue record wins on ownership, because a drone cannot observe who owns a plot. Conflating the two produces a system that will overwrite a title with a photograph.

9. Evaluation and results

Trained on one synthetic city, evaluated on a second with a different layout and a different damage profile. Full pipeline 45.6 s on one core; 3,047 legacy parcels against 2,952 reference parcels over 52,468 candidate pairs.

Stage	Measure	Result
Georeferencing	positional RMSE	8.4529 m → 3.127 m
Georeferencing	inliers in the robust fit	2,133
Schema inference	columns mapped	7/7
Schema inference	area unit, inferred from geometry	bigha at 0.9672
Blocking	true pairs retained	1.0
Matching	precision / recall / F1	0.999 / 0.9997 / 0.9993
Matching	ROC AUC	1.0
Matching	share of decision from geometry	95.6%
Confidence	expected calibration error	0.00009
Assignment	1:1 / subdivision / amalgamation	2,791 / 148 / 5
Topology	total errors	6,168 → 215
Topology	overlapping parcel pairs	5,038 → 0
Topology	doubly-claimed land	18,038.6 → 0.0 m ²
Restraint	parcel-sized gaps refused, not filled	69 of 73 deleted
Change	new / demolished / extended / heightened	31 / 28 / 70 / 59
Encroachment	sites on public land	9, 251.8 m ²
Targeting	predicted / achieved / random	1.4479 / 1.3177 / 2.0567 m

Two results that deserve explanation

The geometric share is reported because it is a falsifiable claim about mechanism. An early version of this

model scored F1 0.9955 by learning to join on khasra number and owner name, with IoU contributing 0.24% of the decision. It would have been useless in any jurisdiction that renumbers on resurvey. The harness now renumbers 45% of blocks and genuinely changes 12% of owners, and a test asserts that geometry accounts for at least 70% of the decision. It currently accounts for 95.6%.

The area ledger reconciles land rather than reporting a total. Summed parcel area falls by 2.879% across topology repair, which looks alarming until it is decomposed: doubly-claimed land falls from 18,038.6 to 0.0 m², while the union footprint — ground counted once — *rises*. The summed figure falls because land recorded as owned by two parties simultaneously is now counted once. No land was created or destroyed, and the ledger demonstrates it.

10. Limitations, stated plainly

- The cadastral parcel layer is **synthetic**, derived from OpenStreetMap block structure and labelled as such throughout the product and its exports. No bulk Indian urban cadastre is publicly available. Building footprints, the road network and the terrain are real.
- Encroachment cases are **deliberately injected**: the generator never places structures on public land, so without injection there would be nothing to detect and the claim would be untested.
- The survey-plan prediction is **calibrated, not exact**. A scale factor measured on one city transfers to a held-out city within roughly 12% and errs conservative.
- Per-parcel digitising noise is **irreducible** by survey control. The system reports that floor rather than promising accuracy below it — a limitation shared with [2], whose displacement model is likewise smooth.
- The system detects and evidences change. It does **not** produce a mutation record, which is a statutory act requiring notice and hearing and whose form varies by state.
- Redaction of personal data is enforced server-side by the API. In the static demonstration build every artifact is delivered to the browser, so redaction there is a display control only. This distinction is documented in the repository.

11. Open questions we would take further

- **Sliver classification with legal consequence.** Distinguishing a digitisation artefact from a genuine tiny urban parcel is classifiable from area, compactness, khasra presence, RoR presence and footprint presence. We currently classify on geometry alone and flag the rest.
- **Invisible boundaries.** Fusing imagery, legacy geometry, textual record and utility topology to infer a boundary with no physical trace is the ceiling of the vision-only approach and is unaddressed in the literature we found.
- **North/south legal regimes.** Preserving measurement authority where the Field Measurement Book is primary, rather than polygon authority, needs a per-state policy object rather than a global setting.
- **An Indian benchmark.** A small, released, ground-truthed harmonisation benchmark for one pilot ULB would let this field measure itself.

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Appendix A — claims we deliberately do not make

Several figures circulate widely in Indian land-administration discussion without traceable provenance. We exclude the following rather than risk being unable to defend them:

- **“Land disputes take about 20 years to resolve.”** Attributed to NITI Aayog across many secondary sources; we could not open the original. [S]
- **“324 years to clear the case backlog.”** Source not located. [U]
- **“25% of Supreme Court decided cases involve land disputes.”** Widely repeated, original study not found. [U]
- **Any figure for man-hours or rupee cost of manual GIS harmonisation** specifically, as distinct from survey or digitisation cost. We searched and found nothing credible in the public domain, and we do not invent one. Where a cost proxy is needed we use the ₹56,725 per sq km resurvey rate [15], which is official.
- **“Indian Geodetic Datum 2023.”** This does not exist. The National Geospatial Policy 2022 [7] commits to *redefining* the national geodetic framework; the Indian Vertical Datum is real and is referenced in [6].
- **The exact 14-character composition of ULPIN**, and the contents of the NAKSHA SDMS schema, whose SOP is available only as a scan without a text layer. [U]

Sources [1]–[4] and [8] were read in the original for this document. Sources [5]–[7] and [9]–[17] are drawn from a sourced research dossier compiled for this project; figures attributed to them carry their grade, and any figure intended for publication should be re-checked against the primary document.